

The Price of Risk: Statistical Learning Approaches to Loan Interest Rate Prediction

MA 5755 — Data Analysis & Visualization · Project - 2026 · Initial Report

Team Details

Name: Aswini V J **Roll No:** MA25M007 **Working alone.**

Dataset Description

Source: LendingClub Loan Data 2007–2020, Kaggle — <https://www.kaggle.com/datasets/ethon0426/lending-club-20072020q1>

Layman Description: LendingClub is a peer-to-peer lending platform that assigns each borrower an interest rate reflecting their credit risk. This project models what borrower and loan characteristics drive that rate, using the platform's full loan history from 2007–2020.

Response Variable: `int_rate` (interest rate, %) — continuous and non-categorical.

Observations & Predictors: ≈ 2.9 million loan records; a stratified sample of 50,000 records will be used for modelling. 19 curated predictors are retained after removing administrative identifiers, post-origination leakage variables (including `grade`, which is derived from the target), and columns with $>30\%$ missing values. Key predictors include: `loan_amnt`, `annual_inc`, `dti`, `fico_range_low`, `revol_util`, `emp_length`, `home_ownership`, `purpose`, `open_acc`, `delinq_2yrs`, `pub_rec`, `term`, `mort_acc`, `installment`, and others.

Research Questions

1. How well do publicly observable borrower financials explain LendingClub's assigned interest rates, and which financial signals carry the most pricing power?
2. To what extent do unsupervised borrower clusters correspond to LendingClub's risk rating tiers, and what financial profiles define each segment?
3. At what point does model complexity stop yielding meaningful gains in predictive accuracy, and what does this imply for model selection in a regulated lending context?

Planned Methods (covered in MA 5755)

1. **EDA & Visualisation:** distributions, correlation heatmaps, outlier analysis, PCA and t-SNE.
 2. **Unsupervised Clustering:** K-means with silhouette evaluation; Gaussian Mixture Models with BIC/AIC comparison (if K-means clusters are weak).
 3. **Regularised Regression:** OLS baseline, Ridge, and Lasso with cross-validated hyperparameter tuning.
 4. **Tree-Based Methods:** Random Forest with feature importance analysis.
 5. **Neural Networks:** MLP with dropout and batch normalisation in PyTorch.
-